Detailed Per-Class Forensic Performance Tables (11 Scanner Classes)

Performance evaluation on the permanently locked 680-sample test manifest across all 11 scanner hardware units:

Hard...	Man...	Opti...	Sup...	Res...	Res...	Res...	Hybr...	Hybr...	Hybr...	Ran...	Ran...	Ran...	SVM ...	SVM ...	SVM ...
Cano...	Cano...	CIS	62	0.98	0.98	0.98	0.68	0.84	0.75	0.58	0.61	0.59	0.32	0.35	0.33
Cano...	Cano...	CIS	62	0.92	0.98	0.95	0.73	0.66	0.69	0.54	0.47	0.5	0.3	0.28	0.29
Cano...	Cano...	CIS	62	1.0	1.0	1.0	0.85	0.73	0.78	0.56	0.53	0.54	0.35	0.32	0.33
Cano...	Cano...	CCD	62	0.95	0.97	0.96	0.82	0.82	0.82	0.57	0.58	0.57	0.33	0.34	0.33
Cano...	Cano...	CCD	60	0.97	0.93	0.95	0.82	0.78	0.8	0.55	0.52	0.53	0.32	0.3	0.31
Epso...	Epso...	CCD	62	1.0	0.95	0.98	0.93	0.87	0.9	0.62	0.65	0.63	0.37	0.38	0.37
Epso...	Epso...	CCD	62	0.95	1.0	0.98	0.87	0.94	0.9	0.6	0.63	0.61	0.35	0.36	0.35
Epso...	Epso...	CIS	62	0.98	0.95	0.97	0.76	0.61	0.68	0.52	0.5	0.51	0.32	0.3	0.31
Epso...	Epso...	CIS	62	1.0	0.97	0.98	0.65	0.81	0.72	0.54	0.52	0.53	0.3	0.28	0.29
Epso...	Epso...	CCD	62	0.95	1.0	0.98	1.0	1.0	1.0	0.78	0.76	0.77	0.46	0.44	0.45
HP S...	HP	CIS	62	1.0	0.97	0.98	1.0	1.0	1.0	0.84	0.82	0.83	0.45	0.42	0.43

Summary Macro Averages:

- **Deep ResNet-18 Baseline:** Accuracy = **97.35%**, Macro-Precision = **0.9741**, Macro-Recall = **0.9735**, Macro-F1 = **0.9735**
- **Dual-Branch Hybrid CNN:** Accuracy = **82.35%**, Macro-Precision = **0.8250**, Macro-Recall = **0.8235**, Macro-F1 = **0.8231**
- **Random Forest Baseline:** Accuracy = **58.53%**, Macro-Precision = **0.6012**, Macro-Recall = **0.5853**, Macro-F1 = **0.5787**
- **SVM (RBF) Baseline:** Accuracy = **33.97%**, Macro-Precision = **0.3245**, Macro-Recall = **0.3397**, Macro-F1 = **0.2828**